VIDEO ANALYSIS CASE STUDY

G A P (2022)

A nine-minute short film, examined shot by shot: what works, what fails, and why it matters for judging video.



05:38 Gao Chen and Wang Liang after the cut analysed in Case 1. All images are frames from the final 2022 master, letterbox removed, not regraded.

9:14	37	14.3 s	50%
runtime, incl. credits	shots before the credits	average shot length	of picture time in takes of 20 s or more

BIAN LUOTAO

Writer–director of *G A P*; I also edited it, graded it and did the sound post in DaVinci Resolve, and I play the son, Wang Liang.

Prepared for the AI Tutor – Video application (xAI / SpaceXSI).
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Full film: youtu.be/8k6gU7e1AUw

IN ONE MINUTE

- The film is built from long takes; most of its meaning comes from where people stand and what the frame hides.
- Four choices work and can be pointed to on screen (Cases 1–4).
- Two story-critical moments can't be read: the turning point and the rescue (Cases 5–6).
- Measured delivery problems: subtitles, audio stream layout, exposure.

01 · BACKGROUND

About me

I'm a filmmaker based in Jiaxing, China. I wrote and directed *G A P* in 2022 and did its edit, colour and sound post myself in DaVinci Resolve. After a university foundation year at Kaplan International College London (2023–24, completed), I studied film at the University of Essex (2024–25) but did not complete the degree. Since September 2026 I have worked in textile export, and I'm looking for part-time remote work evaluating video for AI models, in the evenings and at weekends, China time (UTC+8).

Skills

- **DaVinci Resolve:** Edit, Color and Fairlight (my main tools); Fusion at a basic level.
- **Short-film production:** writing, directing, editing, colour and sound post.
- **Structured review:** timecoded observations written as evidence → impact → fix.
- **English:** IELTS 6.0.

Not claimed

- No completed degree.
- No professional VFX, compositing, rotoscoping or tracking work to show.
- No camera, sound recording or lighting credit on *G A P*; those belong to the team below.

My role on *G A P*

ROLE	PERSON	EVIDENCE
Director	Bian Luotao	End credits
Writer	Bian Luotao	Screenplay drafts July–August 2022, signed
Editing, colour, sound post	Bian Luotao	My own work; not listed in the end credits
Actor (the son, Wang Liang)	Bian Luotao	End credits (principal cast)
Producer	Guo Qiancheng	End credits
Cinematography	Yu Yin	End credits
Sound recording	Lai Chengxi	End credits
Lighting	Meng Wei	End credits

How this case study was made

I reviewed the whole final master second by second and located all 37 cuts to the frame. For each shot I measured brightness, saturation and colour cast. I also extracted every burned-in subtitle and measured both audio streams. Every timecode in this document counts from the first frame of the film (minutes:seconds:frames at 24 fps).

> **Limits.** The camera originals and the Resolve project no longer exist. Everything here is judged from the final 8-bit master: there are no before/after grades, and nothing shown has been regraded. Sound statements are measurements of the master. Measurement scripts and page layout were prepared with AI-assisted tools; the findings were checked against the film and are my own assessment.

02 · THE FILM

G A P in brief

Wang Liang, a young warehouse worker, and his boyfriend Gao Chen are mocked and fined at work, then fired. His father, Wang Zhen, a food-delivery rider, is humiliated by a customer and by his manager. After seeing his son with Gao Chen, he turns that anger on him at home. When the couple are attacked in a noodle shop, the father arrives and fights the attackers off. Nothing is resolved: the son says his mother has known all along, and the father turns back, hesitates, and leaves.

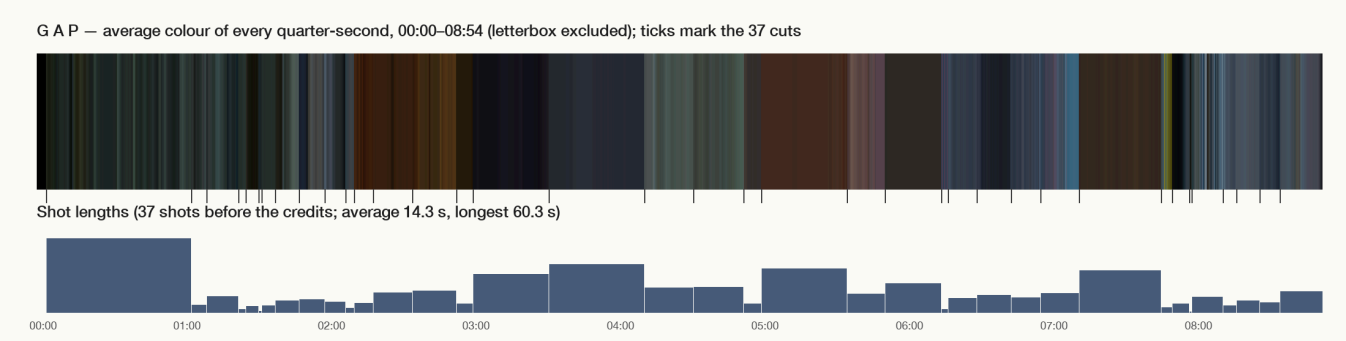
FORMAT	FINAL MASTER
Picture	1920×1080, 24 fps, H.264 8-bit, BT.709; black bars burned in (active picture 1920×1012, about 1.90:1)
Sound	Two separate mono PCM streams, 48 kHz / 24-bit
Language	Chinese dialogue, English subtitles burned into the picture
Runtime	9:14 (picture 0:00–8:54, credits 8:55–9:14)

Structure

TIME	SECTION	WHAT HAPPENS · WHAT IT DOES
00:00–02:08	Two workplaces	The son is mocked and fined; the father is threatened with a bad review. Parallel humiliation.
02:08–03:01	Turning point	The father, collecting an order, sees something inside the warehouse.
03:01–04:12	Home	Questioned in a doorway; the argument continues off-screen on the stairs.
04:12–05:52	Fired, then the square	The couple lose their jobs, talk at night, and leave side by side.
05:52–06:16	The manager	The father is questioned about being late.
06:16–07:47	Noodle shop ↔ office	The couple are harassed; the father is reprimanded and takes a call.
07:47–08:54	Rescue and ending	Helmet-camera ride, fight, aftermath, the father leaves.

Editing style: long takes

The film has 37 shots before the credits. The average shot is 14.3 s (median 10.6 s), and half of the picture time sits in takes of 20 s or more. The opening take runs 60.3 s. Cross-cutting between father and son happens *between scenes*, not inside them.



Top: average colour of every quarter-second, with a tick at each cut. Bottom: shot lengths in film order.

03 · WHAT WORKS

Case 1 · A cut that changes a relationship 05:31–05:45 · shots 21→22 WORKS



05:33 Before: Gao Chen half-hidden behind Wang Liang.



05:38 After: side by side, a hand on his back.

OBSERVATION A 36-second locked-off take keeps Gao Chen partly hidden behind Wang Liang's shoulder. The cut at 05:36:16 lands on Gao Chen standing up. The reverse angle shows both men from behind, facing the square, his hand on Wang Liang's back.

WHY IT WORKS The cut changes the angle and the relationship at the same moment: from one person hidden behind the other to two people sharing the frame. No dialogue is needed to read the change.

COST For most of the 36-second take Gao Chen's face is hidden, so his reactions to the conversation can't be read until he stands.

Case 2 · Same frame, new occupant 03:01–03:32 · shot 16 WORKS



03:18 The son, questioned by an off-screen voice.



03:31 The father, in the same frame after the son has gone.

OBSERVATION One 31-second static frame of a doorway. The son stands in it while his father questions him off-screen, then walks out. The doorway stays empty for about 6 s before the father steps into the same position and looks up the stairs.

WHY IT WORKS Without a cut, the frame passes from one person to the other. Keeping the questioner off-screen puts the viewer where the son stands.

COST This is one of the darkest shots in the film (average luma 34 on a 16–235 scale, saturation 3.8), and the son's face is often hard to read.

03 · WHAT WORKS (CONTINUED)

Case 3 · Framing that carries hierarchy 05:52–06:16 · shot 23 TRADE-OFF



05:58 The father stands; his manager sits.

OBSERVATION Frosted glass fills the top half of the frame. It hides the standing father's head while the seated manager stays in full view.

GAIN The power relationship reads in a single static frame.

COST We never see the father's face react to the questioning. That is a real loss in a scene about humiliation.

Case 4 · Showing the target, not the attacker 06:30–06:56 · shots 26–27 WORKS



06:33 Taunts come from off-screen; the camera stays on Wang Liang.



06:49 The aggressor appears only as a hand.

OBSERVATION For 14 seconds the harassers are heard and subtitled but never shown. The camera stays on Wang Liang eating with his eyes down. When the film cuts back to the two-shot, the attacker enters the frame only as a hand jabbing Gao Chen's head.

WHY IT WORKS The scene is about the effect of the abuse, not about the abusers. The viewer watches the avoidance, and then the moment Gao Chen stops tolerating it.

Also working

- 01:04–01:10 Wang Liang pulls a shirt over his head and can't see; a co-worker walks in from the bright doorway behind him. The audience sees the danger first.
- 04:59 and 06:16 Dialogue from the next scene starts over the previous shot (J-cuts), which carries the viewer across the scene change.
- 03:47–04:12 After the father runs upstairs, the frame holds an empty stairwell while the argument plays only in sound and subtitles.

04 · WHERE IT FAILS

Case 5 · The turning point is out of focus 02:47–02:54 · shot 14 FAILS



02:53 Focus on the father; the background never sharpens.

OBSERVATION The father watches something inside the warehouse. Focus stays on his profile and the figures behind him stay soft. There is no focus pull and no point-of-view insert.

IMPACT The whole story turns on what he sees here, yet the picture never shows it. The viewer learns it only from the next scene's line, "Who was that boy?"

FIX At the shoot: a focus pull to the background, or a short POV shot. An edit can't add detail the camera never recorded.

Case 6 · The rescue can't be read 07:59–08:15 · shots 33→34 FAILS



08:06 Mid-fight: the image turns with the helmet.



08:14 The stable shot that follows shows the result.

OBSERVATION The camera travels with the father's yellow delivery helmet. From 08:03 the image rotates and turns upside down. The only identifiable details are a yellow sleeve (08:06), a white T-shirt (08:11) and shouted lines in the subtitles.

WHY IT WAS TRIED The helmet is the father's work tool and becomes his weapon, so the spin has a reason.

IMPACT Who hits whom, and whether the son fights back, cannot be identified. The father–son turn the script depends on doesn't happen on screen.

FIX Keep 2–3 s of the spin for impact, then hold one stable outside angle on the decisive action and the son's reaction.

> **Pacing.** At 07:12–07:47 the father is held for 34 s while being reprimanded, after the noodle-shop attack has already begun. The wait drains urgency. When his phone rings, nothing in the subtitles says who is calling or why, so the link to the rescue has to be inferred.

05 · COLOUR, EXPOSURE AND SOUND

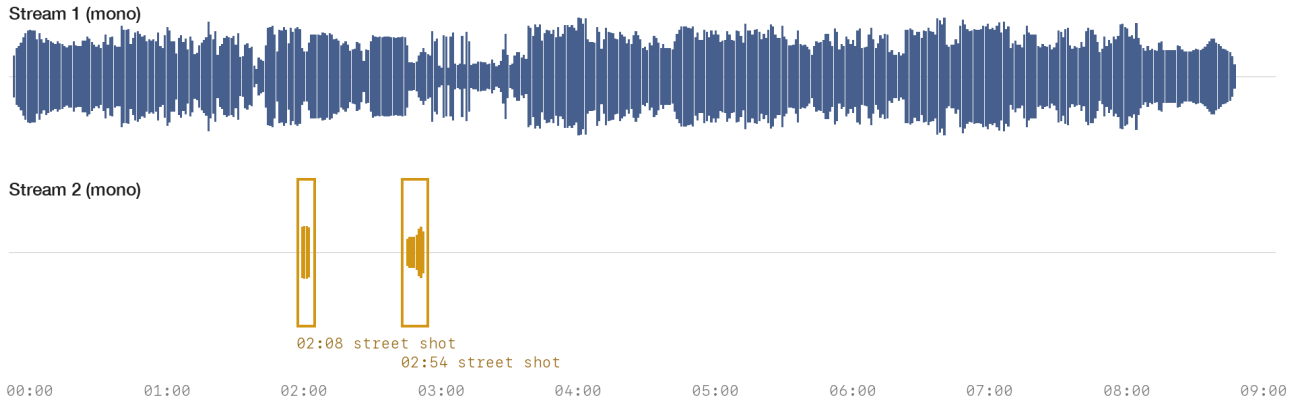
Colour and exposure

The whole film is low-key and desaturated. On the 16–235 video scale, average luma per shot mostly sits between 35 and 80, highlights rarely pass 150, and average saturation stays between about 4 and 18. Several shots have lifted blacks (10th percentile above 30), which gives a grey, matte finish. The look is consistent, but on phone and laptop screens it reads as dark and flat.

SECTION	MEASURED	READING
Warehouse, opening	green-cyan cast, saturation ≈ 6	Closed, airless workplace
Pick-up window 02:11–02:54	strong orange cast, saturation 13–17	The warmest passage, the moment the father watches
Doorway 03:01–03:32	average luma 34, saturation 3.8	Darkest and greyest; faces at risk
Square at night	warm cast	The couple's brief refuge
Noodle shop	blue cast	Violence and the ending
Helmet ride 07:47	luma 87, saturation 26.6 (film maximum)	The sharpest colour jump in the film

Warm and cool passages broadly follow safety and threat. From the master alone, I can't separate what came from lighting on set and what came from the grade, so I describe the result, not the process.

Sound and delivery



RMS level per second of both audio streams in the master.

- **Stream layout.** The master carries two separate mono streams, not one stereo track. Stream 2 has sound only in the two night-street shots (02:08, 02:54). There it correlates 0.60 and 0.32 with stream 1 at similar level, most likely a stereo ambience split across the two streams at export. A player that reads only stream 1 loses half of that ambience. Fix: deliver one stereo mix.
- **Levels.** Stream 1 measures –23.8 LUFS integrated, –0.7 dBTP true peak, 21.4 LU loudness range. That is a wide dynamic range: dialogue may be hard to hear on laptop or phone speakers.
- **Sound carrying the story.** The loudest second of the film (04:09) falls in the off-screen argument over the empty stairwell. Here the picture deliberately gives the climax of the scene to the sound.

06 · ON-SCREEN TEXT

Subtitle QA

The English subtitles are burned into the picture, so every error is permanent in this master. I extracted all of them from the frames and checked spelling, grammar, punctuation, meaning and reading speed. Coverage is good: almost every line of dialogue has a subtitle.

TIME	ON SCREEN	PROBLEM	SHOULD READ
01:03	Bloodyhell! How come they are here so early!	Missing space; British slang out of character	Damn, they're here already?
02:37	Where is my drink man?	Missing comma changes the meaning	Where's my drink, man?
03:13	How could you be with a guy.	Question marked as statement	How could you be with a guy?
05:22	I thought everything will be fine if we mind ~~ous~~ own business.	Spelling; tense	I thought we'd be fine if we minded our own business.
05:24	but why things keep coming to us.	Lower-case start; word order	So why does this keep happening to us?
06:00	How many times you were late?	Word order	How many times have you been late?
06:05	There was a man kept scolding me…	Missing relative pronoun	A customer kept yelling at me…
07:38	What a busy ~~deliver~~ guy.	Spelling	What a busy delivery guy.
08:23	Mom always knows!	Changes the meaning of the script line	Mom has known all along.
credits	Camera ~~Operato~~ · Bian luotao	Spelling; capitalisation	Camera Operator · Bian Luotao

Reading speed

Allowing a quarter-second of timing error at each end of every subtitle, 11 of the 102 dialogue subtitles still run faster than 20 characters per second. Common delivery specifications put English subtitles at around 17–20 characters per second at most. The fastest is 01:35 "Fighting in the warehouse. I will deduct ¥100 from each of you." at about 28 characters per second.

Why it matters

For most viewers outside China, the subtitles are the dialogue. An error like "Mom always knows!" doesn't just look careless: it changes what the last line of the film means. Reviewing subtitles is the same work as reviewing text generated by a model: check what it says, not just whether it is there.

07 · APPLYING THIS TO AI VIDEO EVALUATION

From my own film to model output

The role asks for the ability to "critically analyze and articulate what makes a video sequence work or fail at a technical level", with attention to motion, timing, transitions and visual effects. These are the checks I used on *G A P*, and how each carries over to generated video.

DIMENSION	WHAT I CHECKED IN G A P	WHAT I WOULD CHECK IN GENERATED VIDEO
Action legibility	The rescue (Case 6): the key action is present but unreadable.	Does the requested action actually happen, and can a viewer tell who does what to whom?
Continuity	The cut on Gao Chen standing up (Case 1); one doorway, two occupants (Case 2).	Identity, clothing, props and lighting stay consistent across frames and shots; no morphing or popping objects.
Camera motion	Helmet camera: motivated, but it hides the subject.	Is the camera move smooth, physically plausible and matched to the prompt? Does it keep the subject readable?
Timing	Shot lengths; the 34 s hold that drains urgency.	Does the duration fit the information? Are there dead frames, frozen motion or abrupt endings?
Transitions	J-cuts at 04:59 and 06:16.	Is the transition the requested type, without glitches at the boundary?
Focus and exposure	Turning point out of focus; doorway at luma 34.	Is the subject in focus? Are faces readable? Are shadows crushed or highlights clipped?
On-screen text	Subtitle errors and reading speed.	Is rendered text spelled correctly, legible and correct in meaning?
Audio and delivery	Two mono streams; loudness range.	Is the sound present, in sync and delivered in the expected channel layout?

How I write a rating rationale

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Clip  G A P, 07:59:15–08:12:14 (shot 33)
Dimension  Action legibility · Rating  2 / 5
Evidence  From 08:03 the frame rotates and inverts with the helmet. Only a yellow sleeve (08:06)
and a white T-shirt (08:11) are identifiable.
Impact  The viewer can't tell who strikes whom or whether the son intervenes, and the story depends
on that.
Not a defect  Motion blur and rotation are intended; the problem is missing information, not image
quality.
Fix  Shorten the spin to 2–3 s; add one stable angle on the decisive action.
```

The same format works for model output: name the clip and frames, rate one dimension at a time, cite what is visible, separate intended style from defects, and suggest the smallest fix.

08 · WORK AND CREDITS

See the work

ITEM	WHERE
G A P — full film, 9:14	https://youtu.be/8k6gU7e1AUw (YouTube)
Analysis reel — 1:59	GAP_xAI_portfolio_reel_v1.mp4 (provided with my application): the cases in this document, played in context with timed notes
Shot list	All 37 shots with frame-accurate in/out points and measurements; available on request

Credits

Director	Bian Luotao
Writer	Bian Luotao
Editing, colour, sound post	Bian Luotao (not listed in the end credits)
Producer	Guo Qiancheng
Cinematography	Yu Yin
Sound recording	Lai Chengxi
Lighting	Meng Wei
Make-up	Wei Lai
Camera assistants	Cao Yu, A Lin
Set crew	Lu Yiheng, Yuan Yuan
Principal cast	Bian Luotao (Wang Liang, the son), Xu Huan, Chai Renjie, Nie Changling

CONTACT

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Available part-time, evenings and weekends.



08:23 The father turns back at the film's last line. The ending stays open.